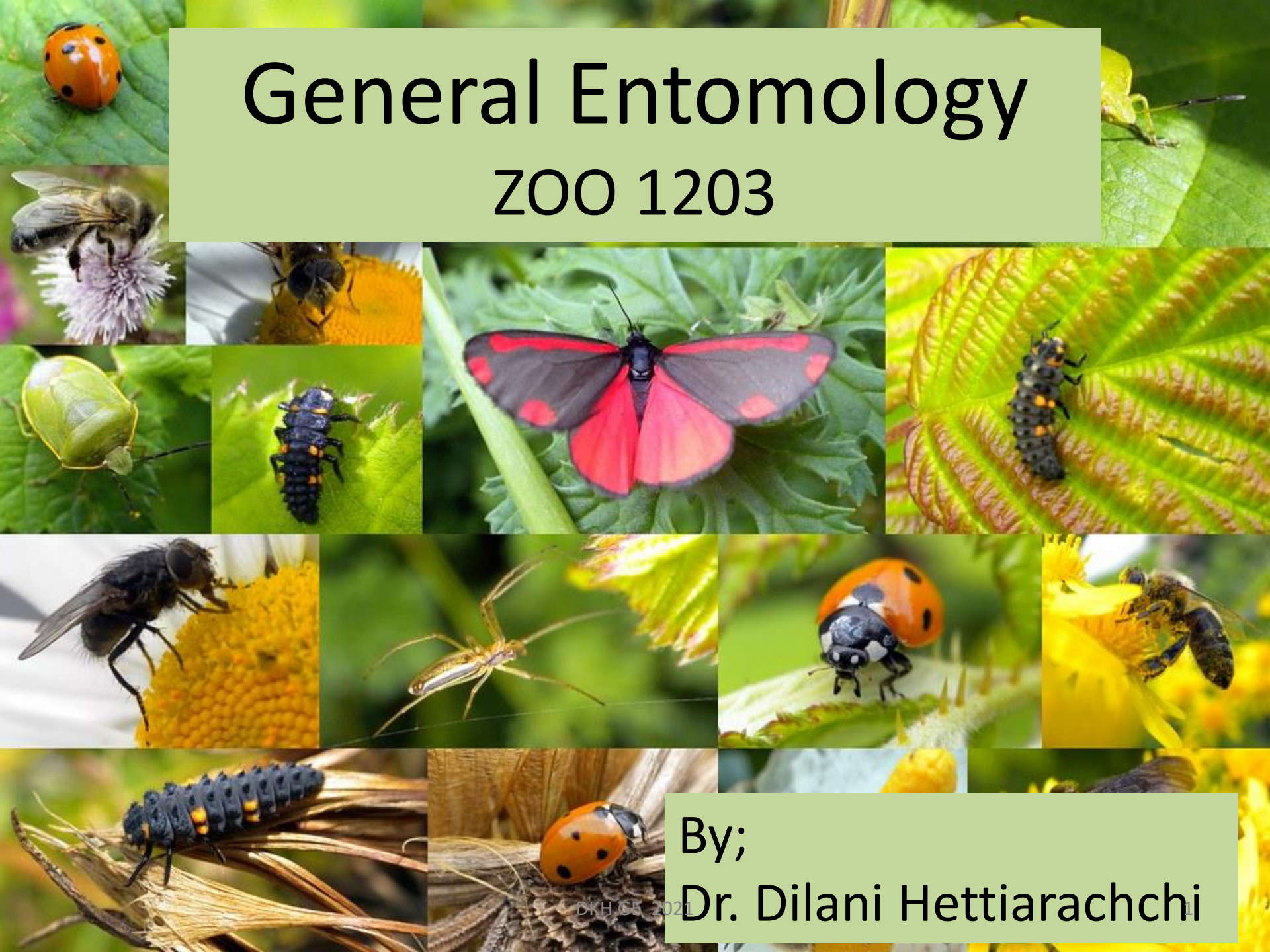


General Entomology

ZOO 1203



By;
Dr. Dilani Hettiarachchi

What is Entomology?

- It is the study of insects
- Entomo → “that which is cut in pieces or segmented”
logy → “study”
- Entomologists → people who study insects, observe, rear and experiment with insects
- Research covers range of biological disciplines like evolution, behaviour, anatomy, physiology, biochemistry, genetics and systematics.

- Entomology is an ancient science
- dating back to the establishment of biology as a formal field of study by Aristotle (384-322 BC).
- There are even earlier references to the use of insects in daily life:
 - such as the growing of silkworms that began 4700 BC in China

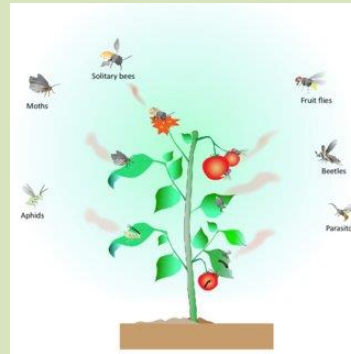


Why study Entomology?

→ Insects are the most abundant group of organism in the world

→ It connects with most other disciplines such as:

- Agriculture
- Biology
- Chemistry
- Food science
- Forensic science and criminology
- Engineering
- Health science



→ Insects are everywhere in the world and you will get in contact no matter where you are

→ Insects are excellent models for physiological and population processes

For example, most used model species for genetic studies is the common fruit fly, *Drosophila melanogaster*

→ More species of insect have had their genome sequenced than any other group of multicellular organisms. Insects are an excellent model for studying the molecular basis of life.

→ Insects have evolved solutions to many physical and chemical problems. Engineers are increasingly looking to insects for solutions in material science and chemistry. The more understanding we have of insects, the more we can put that understanding to use.



→ Insects has inspired man in many ways



Insect themed architecture

Termite-Mound and The Expiatory Temple of the Holy Family Barcelona, Spain by Architect, Antonio Gaudi (1852-1926)



Honeycomb Tires are a marvel of engineering and can't go flat. It doesn't need air therefore it can't go flat – which could be a real lifesaver for members of the military.

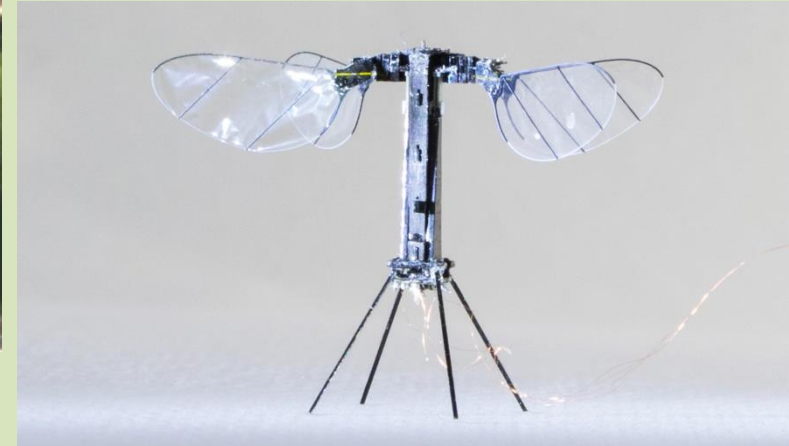


Hive-shaped homes dot the countryside of Ireland and Scotland, dating back centuries. The stone-mosaic hut above blends into the hillside in Kent, Ireland, providing sturdy shelter as well as camouflage. Similar mounds have been found throughout the world, some as old as 4,000 years old. Each region's hive houses were made of different kinds of materials, each blending in to its own landscape. Provide an all-natural heating and air system as well as protection from the elements.

In the 1980s, architect Zvi Hecker was commissioned to design a public housing project in Israel. The result was a topsy-turvy geometric design which is now referred to as the "Honey Bee Hive House." The odd structure in the Ramot Polin neighborhood of Jerusalem is comprised of 720 prefabricated units. It is most certainly one of the most interesting public housing units we've ever seen.



Insect themed robots



Insect themed movies



2014 – bees and others

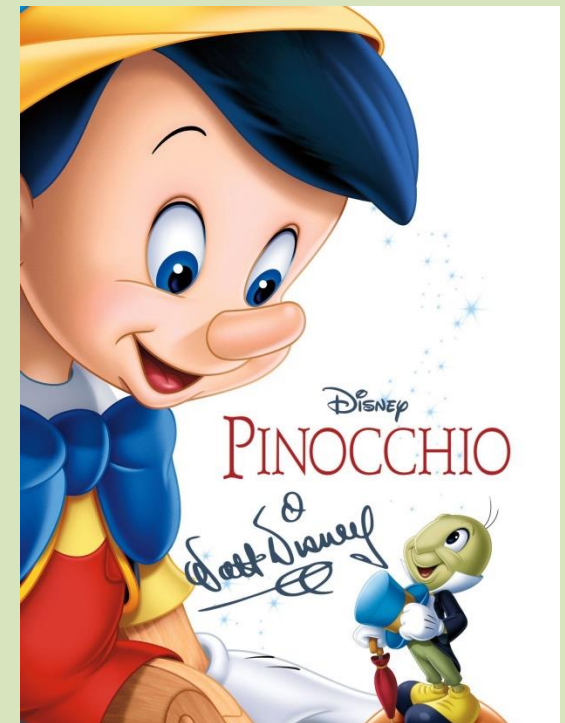
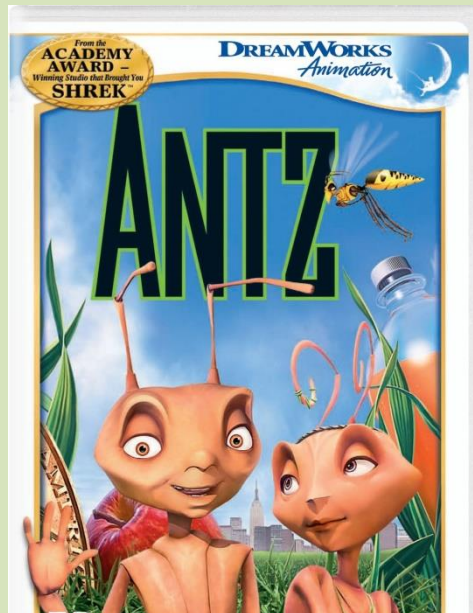
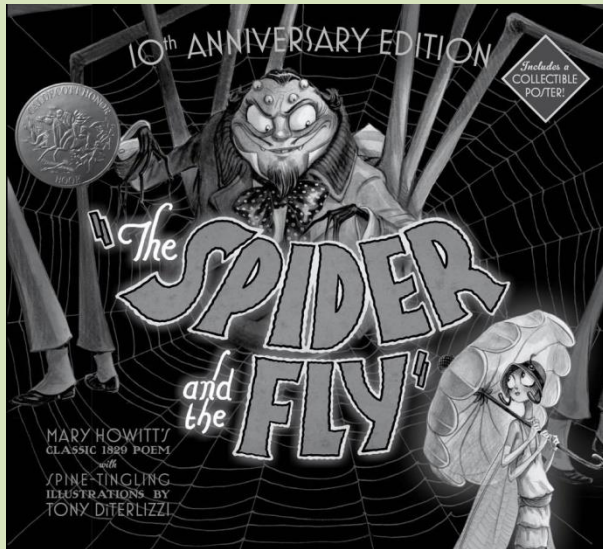
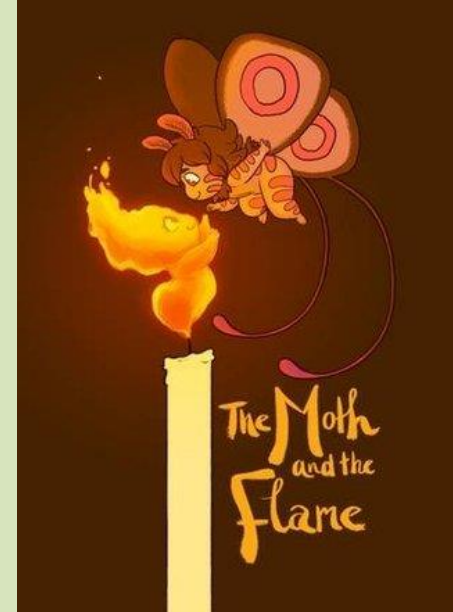


2007 – bees and others

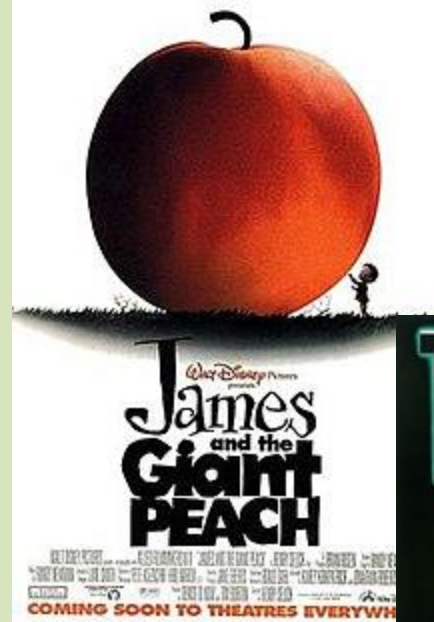


2006 – Ants

- Antz (1998) – Ants and others
- A Bug's Life (1998) – Ants and others
- A Horse Fly Fleas (1947) – Fleas
- An itch in Time (1943) – Flea
- M. Bug Goes to Town (1941) – A bee, grasshopper and other species
- Moth and the Flame (1938) – moths
- The Grasshopper and the Ants (1934)
- Pinocchio (1940) – Cricket
- The Spider and the Fly (1931) - Flies



- James and the Giant Peach (1996)
- The Fly (1986)
- Them!(1954)
- Mimic (1997)
- Honey, I shrunk the kids (1989)
- The deadly mantis (1957)
- Naked lunch (1991)
- The swarm (1978)
- Tarantula (1955)
- Infested (1993)
- The wasp women (1959)
- Bugs (2006)
- They nest (2000)



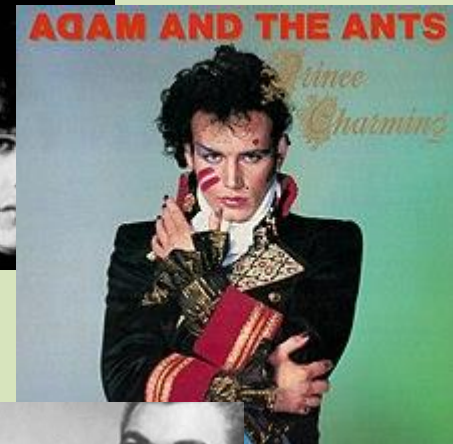
FROM DAVID CRONENBERG, DIRECTOR
OF "THE FLY" AND "DEAD RINGERS"

"ONE OF THE YEAR'S TEN BEST"
-US MAGAZINE -NEW YORK NEWSDAY -BOSTON GLOBE



Bands inspired by insects

- The beetles
- Bee Gees
- Adam and the Antz
- Buddy Holly and the Crickets
- Katydid
- Alien Ant Farm
- Papa Roach
- Iron Butterfly
- WASP
- The Crickets
- The Hives
- Black Moth Super Rainbow
- The Bees
- The Locust
- The Roaches



Abundance and causes of success

- The origin of insects dates back to some 400 million years
- Insects outnumber other existing species
- Insects account for more than two-thirds of all known organisms
- They have many kinds of interactions with humans and other forms of life on earth.



- Their ecologies are incredibly variable
 - Different feeding specializations (detritivory, xylophagy, fugivory, etc..)
 - Can be terrestrial or aquatic
 - Lifestyle maybe solitary, subsocial, or highly social
 - Maybe nocturnal or diurnal
 - Maybe conspicuous or concealed

- Insects often dominate food chains and food webs

- If the insects are eliminated from earth , all the ecosystems will collapse in 4-5 years time.

→ Biologists work with insects for many reasons;

- ease of culturing in laboratory
- rapid population turn over
- availability of many individuals
- minimal ethical concerns compared to vertebrates



Who needs entomologists?

Scientists – By studying insects, entomologists have helped to produce much of our current knowledge on inheritance, physiology, and ecology.

Farmers – Crops and food products are consumed or spoiled by insects. Entomologists are working to sustain optimal agricultural production throughout the world. The work of veterinary entomologists protects livestock from insect-borne illnesses.

Teachers – Teachers use entomology in the classroom to teach basic biological principles and concepts.

Environmentalists – Entomologists study environmental indicators to better understand the relationships between humans and nature. By identifying endangered species and studying their ecosystems, entomologists work to protect the environment and restore threatened habitats.

Industry – Entomologists provide industry with many product opportunities in areas such as biological and chemical pest control. Urban pest control is a \$3 billion industry.

Foresters – Insects are one of the greatest destroyers of timber resources. Entomologists work with foresters to develop effective programs to battle these pests in order to protect forests and their biological wealth.

Animals – Environmentally sound biological control programs developed by entomologists have become standard features in a wide array of solutions which protect animals from harmful chemicals.

The United Nations – Entomologists' research is essential in solving insect-related food shortages and diseases throughout the world.

Customs Inspectors – Entomologists help customs intercept new pests before they enter a country and cause problems.

The Military – Soldiers, marines, and sailors often encounter exotic insects with exotic diseases. Medical and veterinary entomologists lead the way in research to combat insects that carry diseases.

Police – Forensic entomologists use their expertise to help police solve criminal cases.

Homeowners – Urban entomologists provide answers in our war with structural, turf, and ornamental insect pests.

Artists – Entomologists help to preserve the diversity and extraordinary beauty of insects which artists utilize for designs and patterns.

Writers – Insects are an intriguing part of the ecological web. Entomologists can provide unusual twists to ordinary stories. Writers can spin a good tale because insects fascinate.

Doctors – At least one-sixth of the human race currently suffers from insect-carried diseases. Entomologists provide vital information on the treatment and prevention of these ailments.

Our World – Entomologists and their study of insects help us increase the bounty of the planet and preserve its natural beauty.

THANK YOU!

